



Trailhead



[Apex Testing \(/content/learn/modules/apex_testing?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer\)](https://trailhead.salesforce.com/content/learn/modules/apex_testing/apex_testing_triggers?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer)

>
[Test Apex Triggers](#)

Test Apex Triggers

Learning Objectives

After completing this unit, you'll be able to:

- Write a test for a trigger that fires on a single record operation.
- Execute all test methods in a class.

Prerequisites

Let's test a trigger that we worked with in the [Apex Triggers](https://trailhead.salesforce.com/content/learn/modules/apex_triggers) (https://trailhead.salesforce.com/content/learn/modules/apex_triggers) module. If an account record has related opportunities, the `AccountDeletion`  trigger prevents the record's deletion.

If you haven't yet added the `AccountDeletion`  trigger, follow these steps.

1. In the Developer Console, click **File | New | Apex Trigger**.
2. Enter `AccountDeletion`  for the trigger name, and then select **Account** for the sObject. Click **Submit**.
3. Replace the default code with the following.

```
1 trigger AccountDeletion on Account (before delete) {  
2     // Prevent the deletion of accounts if they have related  
3     opportunities.  
4     for(Account a : [SELECT Id FROM Account  
5         WHERE Id IN (SELECT AccountId FROM Opportunity) AND  
6         Id IN :Trigger.old WITH USER_MODE]) {  
7         Trigger.oldMap.get(a.Id).addError('Cannot delete account  
8         with related opportunities.');
```





3. Select **Is Active**.

4. Click **Save**.

[Apex Testing \(/content/learn/modules/apex_testing?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer\)](#)

If your org contains triggers from a previous unit called `AddRelatedRecord` ,

`CalloutTrigger` , OR `HelloWorldTrigger` , disable them. For example, to disable `AddRelatedRecord` trigger:

1. From Setup, search for `Apex Triggers` .
2. On the Apex Triggers page, click **Edit** next to the `AddRelatedRecord` trigger.
3. Deselect **Is Active**.
4. Click **Save**.

To disable the `HelloWorldTrigger` and `CalloutTrigger` triggers, repeat the previous steps.

Add and Run a Unit Test

Before deploying the trigger, write unit tests to perform the actions that fire the trigger and verify expected results.

First, let's start by adding a test method. This test method verifies what the trigger is designed to do (the positive case): preventing an account from being deleted if it has related opportunities.

1. In the Developer Console, click **File | New | Apex Class**.
2. Enter `TestAccountDeletion` for the class name, and then click **OK**.
3. Replace the default class body with the following.

```

1  @IsTest
2  private with sharing class TestAccountDeletion {
3      @IsTest
4      static void TestDeleteAccountWithOneOpportunity() {
5          // Test data setup
6          // Create an account with an opportunity, and then try
7          to delete it
8          Account acct = new Account(Name='Test Account');
9          insert acct;
10         Opportunity opp = new Opportunity(
11             Name=acct.Name + ' Opportunity',
12             StageName='Prospecting',
13             CloseDate=System.today().addMonths(1),
14             AccountId=acct.Id);

```





20

`Test.stopTest();`

[Apex Testing \(/content/learn/modules/apex_testing?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer\)](https://trailhead.salesforce.com/content/learn/modules/apex_testing/apex_testing_triggers?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer)

[Show More](#)

Test Apex Triggers

The test method first sets up a test account with an opportunity. Next, it deletes the test account, which fires the `AccountDeletion` trigger. The test method verifies that the trigger prevented the deletion of the test account by checking the return value of the `Database.delete()` call. The return value is a `Database.DeleteResult` object that contains information about the delete operation. The test method verifies that the deletion was not successful and verifies the error message obtained.

4. To run this test, click **Test | New Run**.
5. Under Test Classes, click **TestAccountDeletion**.
6. To add all the methods in the `TestAccountDeletion` class to the test run, click **Add Selected**.
7. Click **Run**. Find the test result in the Tests tab under the latest run.

The `TestAccountDeletion` test class contains only one test method, which tests for a single account record. Also, this test is for the positive case. Always test for more scenarios to ensure that the trigger works in all cases, including deleting an account without opportunities and bulk account deletions.

Test data is set up inside the test method, which can be time-consuming as you add more test methods. If you have many test methods, put test-data creation in a test utility class and call the utility class from multiple test methods. The next unit shows you how to take advantage of a test utility class and add more test methods.

Tell Me More

The test method contains the `Test.startTest()` and `Test.stopTest()` method pair, which delimits a block of code that gets a fresh set of governor limits. In this test, test-data setup uses two DML statements before the test is performed. To test that Apex code runs within governor limits, isolate data setup's limit usage from your test's. To isolate the data setup process's limit usage, enclose the test call within the `Test.startTest()` and `Test.stopTest()` block. Also use this test block when testing asynchronous Apex. For more information, see [Using Limits, startTest, and stopTest \(https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_testing_tools_start_stop_test.htm\)](https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_testing_tools_start_stop_test.htm).



Resources

[Apex Testing \(/content/learn/modules/apex_testing?](#)

[trailmix_creator_id=srebello7&trailmix_slug=salesforceplatformdeveloper\).](#)

[Apex Developer Guide: Understanding Testing in Apex](#)
([https://developer.salesforce.com/docs/atlas.en-](https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_testing_intro.htm)

Test Apex Triggers

[us.apexcode.meta/apexcode/apex_testing_intro.htm](https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_testing_intro.htm)).

- [Apex Developer Guide: Triggers](#) (https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_triggers.htm).

Get Ready for the Hands-on Challenge

In order to complete the hands-on challenge for this unit, you'll need to create an Apex trigger on the Contact object named `RestrictContactByName`  with the code below:

```
1 trigger RestrictContactByName on Contact (before insert, before
2 update) {
3     //check contacts prior to insert or update for invalid data
4     for(Contact c : Trigger.New) {
5         if(c.LastName == 'INVALIDNAME') {
6             //invalidname is invalid
7             c.AddError('The Last Name "' + c.LastName + '" is not allowed
8 for DML');
9         }
10    }
11 }
```



GET READY

You'll be completing this unit in your own hands-on org. Click **Launch** to get started, or click the name of your org to choose a different one.

YOUR CHALLENGE

Create a Unit Test for a Simple Apex Trigger

Create a simple Apex trigger which blocks inserts and updates to any contact with a last name of 'INVALIDNAME'. You'll copy the code for the



- Code: **Copy from Get Ready for the Hands-on Challenge section above**

Apex Testing (/content/learn/modules/apex_testing?

trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer).

- Name: TestRestrictContactByName copy

Test Apex Triggers

- Goal: 100% test coverage

- Run your test class at least once

Choose hands-on org

Koushik nelluri Salesforce CRM

Created on 5/9/2026



Launch

Check Challenge to Earn 500 Points

Learn

Badges

Trails

Trailmixes

Superbadges

Trailhead Academy

Career Paths

Certifications

Certifications

Maintain Certifications

Verify Certifications

Community

Trailblazer Community

Events

Agentblazers

Salesforce Developers

Salesforce Admins

Code of Conduct

Report Illegal Content
(EU)

Extras

Trailhead Login

Sales Enablement

Trailblazer Store

Salesforce Help

Download Trailhead GO





[Apex Testing \(/content/learn/modules/apex_testing?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer\)](/content/learn/modules/apex_testing?trailmix_creator_id=srebello7&trailmix_slug=salesforce-platform-developer)

Test Apex Triggers